Organisms are classified into Acidophiles, Neutrophiles and Alkalophiles based on their optimum growth pH.

Acidophiles are organisms that grow best at low pH (0.0–5.5) Example: Most fungi, bacteria like *Acidithiobacillus*, Archaebacteria like *Sulfolobus* and *Thermoplasma*.

Neutrophiles are organisms that grow well at an optimum pH between 5.5 and 8.0. Most bacteria and protozoa are neutrophiles.

Organisms that prefer to grow at pH between 8.5-11.5 are called alkalophiles. These microorganisms are typically found in soda lakes and high carbonate soils. Example: *Bacillus firmus.*

3. Water Activity and Osmosis

Water activity, (a_w) is the ratio of vapour pressure of the solution to the vapour pressure of pure water (a_w values vary between 0 and 1). Water activity is inversely related to osmotic pressure. Organisms that can grow in low a_w values are called osmotolerant. Example: *Staphylococcus aureus*.

Only a few organisms are capable of tolerating high salt concentration and still growing optimally in low water activity. Such organisms are called halophiles. Halophiles can grow in 1-15% Sodium chloride (NaCl) concentrations. Organisms that can grow in very salty environments are called extreme halophiles. (They can grow in 15-30%) NaCl concentration. Example: *Halobacterium.*



Crenation:

Shriveling of cytoplasm in the cell is called crenation. This effect helps to preserve some foods.

4. Oxygen

Most of the microorganisms require oxygen for their optimal growth but some of them survive very well in total absence of oxygen and are killed when exposed to air.

Based on their need and tolerance for oxygen, microorganisms are classified into the following types.

(1) **Obligate aerobes** exhibit growth only at full oxygen level (21% O_2 on air) because O_2 is needed for their respiration and metabolic activities Example: *Micrococcus*, most Algae, Fungi and Protozoa.

(2) **Microaerophiles** are aerobes that require oxygen at levels lower than that of air. Example: *Azospirillum, Campylobacter, Treponema*

(3) Obligate anaerobes does not require oxygen for their respiration and growth. This group cannot tolerate O_2 and are killed in its presence. Example: Methanogens, *Clostridium.*

(4) **Aerotolerant anaerobes** can grow in the presence of oxygen though O_2 is not required for their growth. Example: *Streptococcus pyogenes.*

(5). **Facultative anaerobes** can grow either under oxic or anoxic conditions: Example: *Escherichia coli.* (Figure 6.7)

6.6 Measurement of Growth

Different methods are employed for measuring the cell growth of microorganisms. Cell growth is indicated by increase in the number of cells or increase in weight of cell mass. There are direct and indirect methods of measuring microbial growth.

1. Direct Measurements

Total count and viable count are the two methods widely employed to count cell numbers.

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